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# Causal SCX: Causal Inference

Causal SCX — Multi-Expert Audit of Causal Inference

SCX

2026

## Abstract

**SCX**(scx\_causal), 3-SCM (?) Disagreement  $M \geq 2$  (1) **Causal Consensus**(**Causal Consensus Bound**): Hoeffding awaiting,  $M \{\hat{\tau}_m\}_{m=1}^M$  Consistency(ATE)  $\tau^*$ . Proof  $M, \tau^*$ , (**systematic confounding**) (2)  $Z_1, Z_2, \dots, Z_M$   $M$   $U$   $T$   $Y$   $Z_m, D(\hat{\tau}_1, \dots, \hat{\tau}_M)$ . (3)  $\{Z_1, \dots, Z_K\}$ , IV Disagreement  $Q_k$  ( $k$ ). (4) **do-calculus** do Disagreement., SCX  
**Keywords:** do-calculus,

## 1

—, awaiting. **SCX**(scx\_causal)(?) Disagreement .Core—**3-SCM** (Theorem 1, ?)—(SCM)  $M > 3$  **Causal SCX**—Causal Inference. do-calculus SCX. :  $T \in \{0, 1\}$ ,  $Y \in \mathbb{R}$ ,  $X \in \mathbb{R}^p$ .  
ATE  $\tau^* = \mathbb{E}[Y(1) - Y(0)]$ ,  $Y(t)$ .  $M$   $\hat{\tau}_1, \dots, \hat{\tau}_M, \mathcal{M}_m$ .

## 2

### 2.1

**D 1** (SCM).  $SCM \mathcal{M} = \langle U, V, F, P(U) \rangle$   $U, V, F = \{f_v : v \in V\}$   $U$ .  $\mathcal{M}$   $P_{\mathcal{M}}(V)$   $P_{\mathcal{M}}(V \mid \text{do}(t))$ .

**D 2** ( $\mathcal{M}$ -awaiting).  $SCM \mathcal{M}_1, \mathcal{M}_2$  awaiting,  $\mathcal{M}_1 \equiv_{obs} \mathcal{M}_2$ , :  $P_{\mathcal{M}_1}(V) = P_{\mathcal{M}_2}(V)$ .

**Theorem 1 (3-SCM)**: awaiting SCM  $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$ ,  $Q(\text{ATE})$ :

$$Q(\mathcal{M}_1) \neq Q(\mathcal{M}_2) \neq Q(\mathcal{M}_3) \neq Q(\mathcal{M}_1),$$

$Q$ .  $M$ , Disagreement.

### 2.2 Hoeffding

Hoeffding awaiting:  $X_1, \dots, X_M$ ,  $a_i \leq X_i \leq b_i$ ,  $\varepsilon > 0$ ,

$$P \left( \left| \frac{1}{M} \sum_{i=1}^M X_i - \mathbb{E} \left[ \frac{1}{M} \sum_{i=1}^M X_i \right] \right| \geq \varepsilon \right) \leq 2 \exp \left( - \frac{2M^2 \varepsilon^2}{\sum_{i=1}^M (b_i - a_i)^2} \right).$$

Core: —  $P(V)$ , —  $\hat{\tau}_m$

### 3 Causal Consensus

#### 3.1

$M$   $\{\hat{\tau}_m\}_{m=1}^M$   $\mathcal{Z}_m, g_m$   $()$   $\mathcal{H}_m$  ATE :

$$\hat{\tau}_m = \hat{\tau}(\mathcal{Z}_m; g_m; \mathcal{H}_m).$$

(consensus measure):

$$C_M = 1 - \frac{1}{M(M-1)} \sum_{1 \leq i < j \leq M} |\hat{\tau}_i - \hat{\tau}_j|.$$

$C_M = 1()$ ; Disagreement  $C_M \rightarrow 0$ .

#### 3.2

$$\hat{\tau}_m - \tau^* = \underbrace{\text{bias}(\mathcal{Z}_m, \mathcal{H}_m)} + \underbrace{\varepsilon_m}.$$

**D 3.** (*consensus offset*):

$$\delta_M = \frac{1}{M} \sum_{m=1}^M \text{bias}(\mathcal{Z}_m, \mathcal{H}_m).$$

#### 3.3

**T 4** (Causal Consensus — Hoeffding ). (A1)  $\hat{\tau}_m \in [\tau_{\min}, \tau_{\max}]$ ;

(A2)  $\varepsilon_m \in P(V)$  (/

(A3)  $m_0$   $\text{bias}(\mathcal{Z}_{m_0}, \mathcal{H}_{m_0}) = 0$   
 $\varepsilon > 0$ :

$$P \left( \left| \frac{1}{M} \sum_{m=1}^M \hat{\tau}_m - \tau^* - \delta_M \right| \geq \varepsilon \right) \leq 2 \exp \left( - \frac{2M\varepsilon^2}{(\tau_{\max} - \tau_{\min})^2} \right).$$

,  $\delta_M = 0()$ ,  $O(1/\sqrt{M})$   $\tau^*$ .,  $()$ ATE,  $\delta_M$   $\tau^*$  Scale,  $(C_M \rightarrow 1)$  .

*Proof Sketch.*  $\hat{\tau}_m - \tau^* = \text{bias}_m + \varepsilon_m$ .  $\delta_M + \frac{1}{M} \sum_m \varepsilon_m$ . (A2),  $\varepsilon_m \in P(V)$ , Hoeffding Conclusion.  
 Core: Hoeffding  $()$   $\delta_M$ —Causal Inference”awaiting”.  $\square$   $\square$

**Honest Strike / Honest Critique:**

**Honest Strike:** (§??). **Honest Strike:**, .. ?? ,  $\tau^*$ .

**C 5.** ??,  $\delta_M \neq 0$   $|\delta_M| > \frac{\tau_{\max} - \tau_{\min}}{\sqrt{2M}}$ ,  $\{\hat{\tau}_m\}_{m=1}^M$ .

#### 3.4

,  $M$ , (consensus depth):

$$D(M) = \inf \{d \in \mathbb{N} : \exists d\}.$$

$D(M) > M/2$ , — ..

## 4

### 4.1

$M \mathcal{Z}_1, \dots, \mathcal{Z}_M \subseteq X$ . ()  $\mathcal{Z}^*, : \mathcal{Z}^* T Y$ .  $\mathcal{Z}_m \subsetneq \mathcal{Z}^*, \hat{\tau}_m$  ;  $\mathcal{Z}_m \supseteq \mathcal{Z}^*,$ .

**A 6.** :  $\mathcal{Z}_1 \preceq \mathcal{Z}_2 \iff \mathcal{Z}_1 \subseteq \mathcal{Z}_2$ .  $\mathcal{Z}_{\min}^*$ .

**T 7. Disagreement:**

$$\Sigma_{dis} = (|\hat{\tau}_i - \hat{\tau}_j|)_{i,j=1}^M \in \mathbb{R}^{M \times M},$$

$\rho(\Sigma_{dis})$ . :

$$G_m = \frac{\hat{\tau}_m - \hat{\tau}_{m-1}}{|\mathcal{Z}_m| - |\mathcal{Z}_{m-1}|},$$

$|\mathcal{Z}_m|$  ( $|\mathcal{Z}_1| < |\mathcal{Z}_2| < \dots$ ).  $\mathcal{H}_0()$ :

(i)  $\rho(\Sigma_{dis}) \xrightarrow{p} 0$  as  $n \rightarrow \infty$ ;

(ii)  $\sup_m |G_m| \xrightarrow{p} 0$  as  $n \rightarrow \infty$ ;

(iii)  $\hat{\tau}_m : \exists m_0 : \forall m \geq m_0, |\hat{\tau}_m - \tau^*| < |\hat{\tau}_{m-1} - \tau^*|$ .

(i) (ii) .

*Proof Sketch.*  $\mathcal{Z}_m$  ( $\mathcal{Z}_m \supseteq \mathcal{Z}_{\min}^*$ ),  $\hat{\tau}_m \xrightarrow{p} \tau^*, \forall m$ .  $\Sigma_{dis} \xrightarrow{p} 0$   $G_m \xrightarrow{p} 0$ . ,  $\mathcal{Z}_m \not\supseteq \mathcal{Z}_{\min}^*$ , ,  $\hat{\tau}_m \xrightarrow{p} \tau^* + b_m$   $b_m \neq 0$ . ,  $b_m \Sigma_{dis}$ . , Appendix.  $\square$   $\square$

**D 8.**

$$D_{conf} = \frac{\rho(\Sigma_{dis})}{\frac{1}{M} \sum_{m=1}^M \widehat{\mathbb{V}\text{ar}}(\hat{\tau}_m)},$$

$\widehat{\mathbb{V}\text{ar}}(\hat{\tau}_m)$   $\hat{\tau}_m$ .  $\mathcal{H}_0$ ,  $D_{conf}$  Tracy–Widom ().

”Missing”””” . ,  $\rho(\Sigma_{dis})$  **Honest Strike:** —

## 5

### 5.1

$K \geq 2$   $Z_1, \dots, Z_K, Z_k$  :

(IV1) **(Relevance):**  $\mathbb{C}\text{ov}(Z_k, T) \neq 0$ ;

(IV2) **(Exclusion):**  $Z_k \perp\!\!\!\perp Y(t) \mid T, X()$ ;

(IV3) **(Ignorability):**  $Z_k \perp\!\!\!\perp U()$ .

$\hat{\tau}_{\text{IV}}^{(k)} Z_k$ .  $\hat{\tau}_{\text{IV}}^{\text{full}} K$  ,  $\hat{\tau}_{\text{IV}}^{(-k)} \{Z_1, \dots, Z_K\} \setminus \{Z_k\}$ .

## 5.2

**T 9** (IV).  $k$  :

$$Q_k = \frac{\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)}}{\sqrt{\widehat{\text{Var}}(\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)})}}.$$

$Z_k$  (IV1)–(IV3)( $Z_k$ ),

$$Q_k \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

$Z_k$  (IV2),  $Z_k \rightarrow Y$ ,  $\gamma_k \neq 0$ ,

$$Q_k \xrightarrow{d} \mathcal{N}\left(\frac{\gamma_k}{\sqrt{\widehat{\text{Var}}(\dots)}}, 1\right),$$

$\gamma_k$  ScaleGrowth.

*Proof Sketch.* IV,  $\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)} \xrightarrow{d} \tau^*$ .  $Z_k$ ,  $\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)} \xrightarrow{d} \Delta$ .  $\Delta$ .  $\square$   $\square$

**C 10.**

$$Q_{\max} = \max_{k=1, \dots, K} |Q_k|.$$

”,  $Q_{\max} \leq K Q_{\max}$ .

## 5.3 Disagreement

$M$ .  $m$  Verdict  $\mathbf{v}_m \in \{0, 1\}^K$ ,  $v_{m,k} = 1$   $Z_k$ . **Disagreement:**

$$D_{IV} = \frac{1}{M(M-1)} \sum_{i < j} \|\mathbf{v}_i - \mathbf{v}_j\|_1.$$

$D_{IV}$  Disagreement.  $D_{IV} Q_k(k)$ ;  $D_{IV} Q_k$ .

**Honest Strike / Honest Critique:**

**Honest Strike:** IV ( $F < 10$ ),  $Q_k$  **Honest Strike:** ( $\cdot$ ).  $Q_k$  Co

## 6 do-calculus

### 6.1 Pearl

$G$  (DAG),  $X, Y, Z, W$  ( $\cdot$ ). do-calculus ( $\cdot$ ):

- **1( $\cdot$ ):**  $P(y \mid \text{do}(x), z, w) = P(y \mid \text{do}(x), w) \mid G_{\overline{X}} Y \perp\!\!\!\perp Z \mid X, W$ .
- **2( $\cdot$ ):**  $P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), z, w) \mid G_{\overline{X}\overline{Z}} Y \perp\!\!\!\perp Z \mid X, W$ .
- **3( $\cdot$ ):**  $P(y \mid \text{do}(x), \text{do}(z), w) = P(y \mid \text{do}(x), w) \mid G_{\overline{XZ(W)}} Y \perp\!\!\!\perp Z \mid X, W$ .

## 6.2

**T 11** (do-calculus-SCX ).  $\mathcal{E} = \{\mathcal{M}_1, \dots, \mathcal{M}_M\}$   $M$  ,  $\mathcal{Q}$ .

(1)  $1 \ i, j \ \mathcal{Z}_i \neq \mathcal{Z}_j \text{ awaiting } \hat{\tau}_i \approx \hat{\tau}_j, 1 \ G_{\overline{T}} = \mathcal{Z}_i \Delta \mathcal{Z}_j$

(2)  $2 \ G_{\overline{XZ}} = . \text{ scx\_causal Theorem 1}$

(3)  $3 \ Z \text{ do-Disagreement, SCX } D_{IV} \text{ (Disagreement)} \ G_{\overline{XZ(W)}}.$

: **SCX**

*Proof Sketch.* (1)  $(\Rightarrow)$ : do  $r \Rightarrow$  (2)  $(\Leftarrow)$ : Disagreement SCX  $\Rightarrow$  Disagreement, Core Pearl Bareinboim —do-calculus (??)— scx\_causal.  $\square$   $\square$

**C 12** (Disagreement do ). *SCX M Disagreement(  $\nexists$  Consistency  $n \rightarrow \infty$ ), do DAG —*

**Honest Strike / Honest Critique:**

**Honest Strike:** . scx\_causal 3-SCM Core: awaitingDisagreement. **Honest Strike:** do-calculus (?). DAG ; (, c-awaiting). **Honest Strike:** ?? Proof”

## 7

### 7.1 $M$

$M \rightarrow \infty, \mathcal{H} \Pi, :$

$$\bar{\tau}_{\infty} = \int_{\mathcal{M} \in \mathcal{H}} \tau(\mathcal{M}) d\Pi(\mathcal{M}),$$

ATE  $\Pi$ -.  $\tau^*$  ,  $\Pi$  ( ),  $\bar{\tau}_{\infty}$ . SCX Causal Inference (?).

### 7.2 Compactness

?? ? :  $M, M-1$  ( $\rightarrow \max$ ), 1.Hoeffding —  $M-1$  Compactness  $R \in \mathbb{R}^{M \times M}, R_{ij} = \text{Corr}(\hat{\tau}_i, \hat{\tau}_j)$ .

### 7.3

1 :  $C_M, ;$

2 :  $D_{\text{conf}}, ;$

3 : IV,  $\{Q_k\}_{k=1}^K;$

4 **do** Disagreement, ??

5 **Verdict:** , ATE

### 7.4

(1) : .

(2) :  $M$

(3) :  $p \gg n,$

(4) : . (, ) SCX  $g$ -

(5) : (

## 8 Conclusion

**Causal SCX**—Causal Inference, *scx-causal* 3-SCM *M*. Proof do-calculus SCX. Core: , Causal Inference— *M*, ().—— Causal SCX, *M*, —**Consistency**  $\delta_M$ . Conclusion.: , Causal Inference””

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## A ?? Proof

*Proof.* :  $\mathcal{Z}_m \supseteq \mathcal{Z}_{\min}^*$ , (Pearl, 2009, Theorem 3.3.2),  $\mathcal{Z}_m : \hat{\tau}_m \xrightarrow{p} \tau^*$  as  $n \rightarrow \infty$ .  $i, j$ :  $|\hat{\tau}_i - \hat{\tau}_j| \xrightarrow{p} 0$ .  $\Sigma_{\text{dis}}, \rho(\Sigma_{\text{dis}}) \xrightarrow{p} 0$ . ,  $|\mathcal{Z}_m| \geq |\mathcal{Z}_{\min}^*|$ ,  $G_m \xrightarrow{p} 0()$ . :  $\mathcal{Z}_m \not\supseteq \mathcal{Z}_{\min}^*$ ,  $T \leftarrow U \rightarrow Y$   $\mathcal{Z}_m$ . (Imbens & Rubin, 2015, Ch.12):

$$\hat{\tau}_m \xrightarrow{p} \tau^* + b_m, \quad b_m \neq 0,$$

$b_m$  ScaleMissing.  $S_{\text{conf}} = \{m : \mathcal{Z}_m \not\supseteq \mathcal{Z}_{\min}^*\}$  .  $i \in S_{\text{conf}}, j \notin S_{\text{conf}} : |\hat{\tau}_i - \hat{\tau}_j| \xrightarrow{p} |b_i| > 0$ . Gershgorin,  $|S_{\text{conf}}| \geq 1$   $\rho(\Sigma_{\text{dis}}) \min_{i \in S_{\text{conf}}} |b_i|$ .  $\therefore \Sigma_{\text{dis}} \cdot \mathcal{H}_0, O_p(n^{-1/2})$ , Random Matrix(Tracy–Widom, 1996),  $F_\beta(\beta = 1)$ .  $D_{\text{conf}} \mathcal{H}_0$  Tracy–Widom.  $\square$   $\square$



## B ?? Delta

*Proof.*  $T = \alpha_0 + \mathbf{Z}'\boldsymbol{\alpha}_1 + \mathbf{X}'\boldsymbol{\alpha}_2 + \varepsilon$ .  $Y = \beta_0 + \hat{T}\beta_1 + \mathbf{X}'\boldsymbol{\beta}_2 + \eta$ ,  $\hat{\tau}_{\text{IV}} = \hat{\beta}_1$  ATE IV. ,  $\sqrt{n}(\hat{\tau}_{\text{IV}} - \tau^*) \xrightarrow{d} \mathcal{N}(0, \sigma_{\text{IV}}^2)$ .  $\hat{\tau}_{\text{IV}}^{\text{full}} - \hat{\tau}_{\text{IV}}^{(-k)}$ .  $\hat{\tau}_{\text{IV}}^{\text{full}} - \hat{\tau}_{\text{IV}}^{(-k)}$  (influence function):

$$\psi_k(O) = \psi^{\text{full}}(O) - \psi^{(-k)}(O),$$

$\psi^{(\cdot)}$  ,  $\mathbb{E}[\psi_k(O)] = 0$ ,  $Q_k \xrightarrow{d} \mathcal{N}(0, 1)$ .  $Z_k$ ,  $\mathbb{E}[\psi^{\text{full}}(O)] = \gamma_k \cdot h()$ ,  $\mathbb{E}[\psi^{(-k)}(O)] = 0$ ,  $\lambda_k(n) = O(\sqrt{n}\gamma_k)$ . □